#35345 - DCDC 400V/12V

400V

DC-DC400->12V

Prise marinco

Circuit breaker

Battery Starboard

Battery port

Battery Service

Minus 1

Minus 2

Bus bar 12V- Starboard switched

Bus bar 12V- Port switched

Répartiteur

Fusible

Fusible

Port Engine

Starboard engine

Alternator

Alternator

Switch, BusBar and Coupler Box

#32178 - Battery 12V elec (BA)

#28440 - Console X100 (X100/sport classique)

- Nav light
- Anchor light
- Bilge
- horn

Version sport exclu/knight: #39534

#28440 - Console X100 (X100/sport classique)

- Lights
- Shower
- Icebox
- Courtesy lights
- Options

Version sport exclu/knight: #39534

#25208 - KeyBox IMS Electrique X100

- CAN
- Supply 12V
- **Fan (inverter)**
- **Pump (inverter)**
- Search Lights
- **Auto Off**
- **DCDC ignition**
- AUX 2

#32098 - Faisceau PowerSupply X100Elec

400V_IMS Elec *(lien sur drawio)*

Console

Switches

Fuse Box

Switches

Fuse Box

Keybox X100 Elec

DC-DC12->24V

auto off K3

FBA Electric Drive

Bus bar +&- Direct plus

Helm switch or key with remote control switch (exclusive version)

Bus bar +&- Swiched

Bipolar switch IMS

Bus bar +&- IMS

FB IMS

Fusebox Console

ON/OFF IMS

#36098 - FuseBox Electric Drive Sevcon/Tyva X100/Sport (FBA)

- Emergency brake L
- Emergency brake R
- Supply 12V
- Supply battery 400V (hors Tyva)
- **Cmd MC42 (battery & inverter)**
- CAN Drive
- 120 ohms CAN Drive
- Charger (CAN & Supply12V)
- Battery (CAN with GND, 12V & ignition)
- 120 ohms J1939
- CAN J1939
- HVLP (CAN & 12V)
- Inverter R (CAN & 24V)
- Inverter L (CAN & 24V)
- CAN keybox (commuter)
- **Supply 24V**

#27249 - Fusebox IMS Electrique X100

- Supply 12V
- Supply **MD4** (relay)
- (Proemion)
- RTC
- Supply Direct Battery
- Supply **MC42** (relay)
- ON/OFF IMS
- **Auto OFF K3**
- **Télérupteur CMD**
- Télérupteur supply

CAN/Digit/Analog_X100 Elec *(Lien sur Drawio)*

#34365 - Fusebox console X100

- Perm supply
- switched supply
- GPSMAP
- VHF
- Audio
- Supply NMEA
- Bilge auto
- Bilge repartiteur
- Bilge Cmd
- Bilge manuel
- Cigaret lighter
- Spare 1, 2 & 3
- Option RGB

Version Fusebox console option RGB: #34365 + #34798

Faisceau option:

#22368 - Guindeau Ancre X100

#36090 - Faisceau Toilette x100

#37489 - Fusion BB100

#22437 - Faisceau table

#34798 - Fusebox RGB X100

Lien avec un plan faisceau (RapidHarness)

400V

12V

GND

230V AC

Électrique

12V & 24V (hors Mercury)

Type de bateau: X100 **Electrique**

Author: Arthur Echivard

Date: 05/11/2025

indice: ind B

#25208 - KeyBox IMS Electrique X100

- CAN
- Supply 12V
- **Fan (inverter)**
- **Pump (inverter)**
- Search Lights
- **Auto Off**
- **DCDC ignition**
- AUX 2

#33530 - MD4 - X100

- Buzzer
- **ID TAG 1**
- Joystick_Left
- Joystick_Right
- Supply MD4 12V
- **RTC**
- CAN Main
- G12
- Wifi
- CAN Garmin **(commuter)**

#25844 - BMW charger harness

#39321 - Faisceau Tyva

#35345 - DCDC 400V/12V

IMS

Console

Garmin

Commande IMS

Cooling pump inverter

FAN mk E

DC-DC12V

Keybox X100 Elec

Auto OFF

FBA

Télérupteur IMS

BUS CAN Drive

MD4

ON/OFF IMS

BUS CAN Main

ID Tag 0

G12

#28580 - MC42_X100 Electric

Battery 400V

FBA

BUS CAN Drive

J1939

MC42

230V AC charger

400V_IMS Elec *(lien sur drawio)*

FBA

IMS ON

IMS ON

Oil level

Sevcon Gen4 inverter

Sevcon Gen4 inverter

Sevcon HVLP inverter (en J1939 ou en CAN drive)

BUS CAN Drive

J1939

Battery relay

Inverter relay

FBA "Electric drive"

12V-X100 Elec *(lien sur drawio)*

#29775 - Engine_track_encoder_controler

#29175 - Engine_pump_encoder_controler HVLP

- EV By pass MC41
- EV Down MC42
- EV unlock MC43
- EV lock MC59
- EV tension MC58
- EV up MC57

Locking pressure

Track pressure

Manifold block

Fusebox Console

Fusible 3A

Alim NMEA2000

NMEA 2000

VHF

Autres accessoires Garmin

Transpondeur AIS800

Digital

Analogique

CAN/NMEA2000

12V

400V

Water

230V AC

Lien avec un plan faisceau (RapidHarness)

CAN/Digital/Analogique
Type de bateau: X100 **Electrique**
Author: Arthur Echivard
Date: 05/11/2025
indice: ind A

Mercury

Console







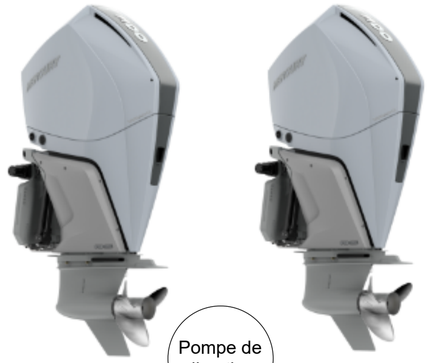






NMEA 2000

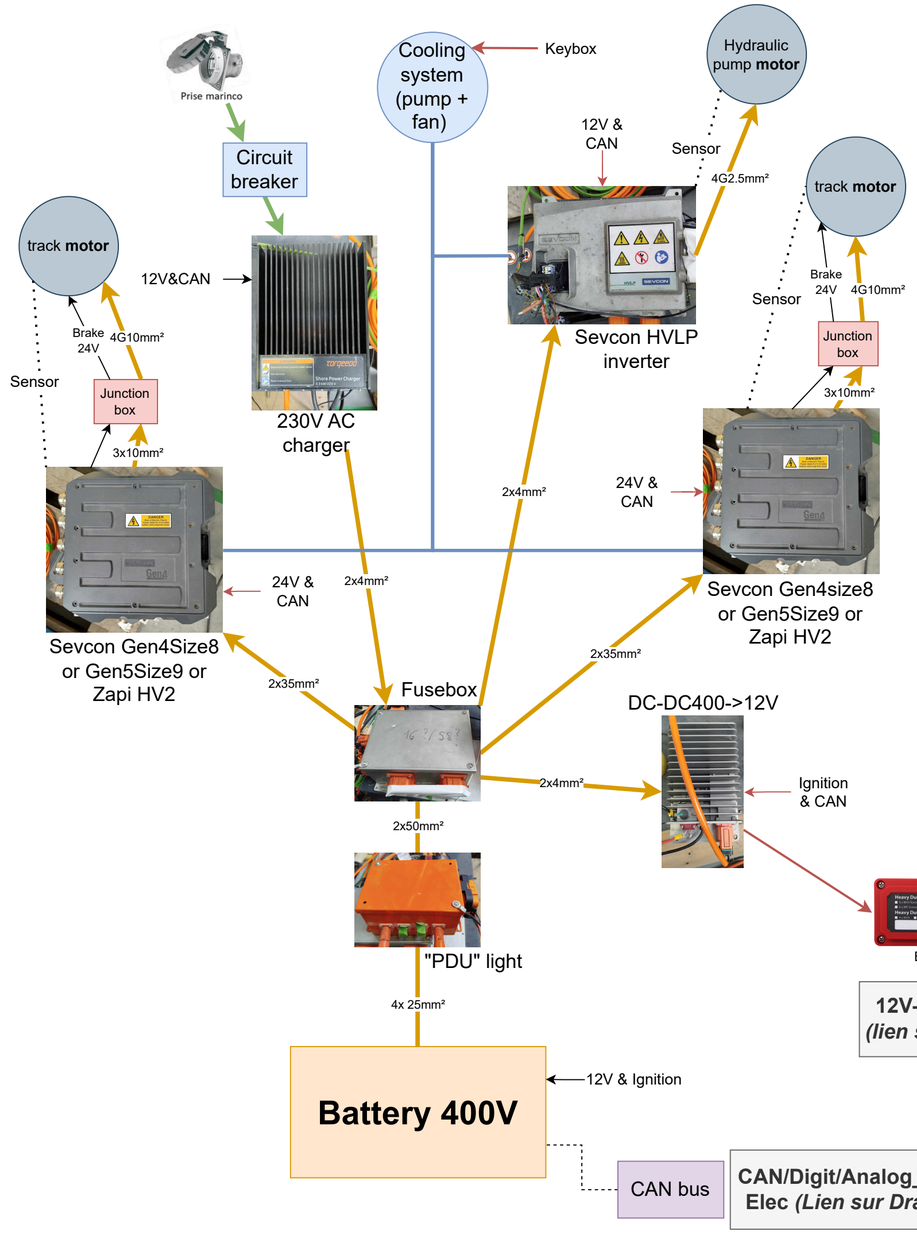




Pompe de direction

Humphree

Bluenav



Data

400V

12V

Water

230V AC

400V
Type de bateau: X100 **Electrique**
Author: Arthur Echivard
Date: 05/11/2025
indice: ind A

Prise marinco

Circuit breaker



#28440 - Console X100 (X100/sport classique)

- Nav light
- Anchor light
- Bilge
- horn

version sport exclu/knight: #39534

#28440 - Console X100 (X100/sport classique)

- Lights
- Shower
- Icebox
- Courtesy lights
- Options

version sport exclu/knight: #39534

#30940 - Faisceau PowerSupply X100

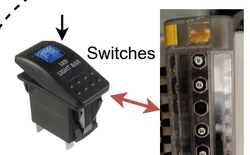
Console

CAN/Digit/Analog_X100 Therm *(Lien sur Drawio)*

#34365 - Fusebox console X100

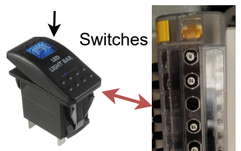
- Perm supply
- switched supply
- GPSMAP
- VHF
- Audio
- Supply NMEA
- Bilge auto
- Bilge repartiteur
- Bilge Cmd
- Bilge manuel
- Cigaret lighter
- Spare 1, 2 & 3
- Option RGB

Version Fusebox console option RGB: #34365 + #34798



Switches

Fuse Box



Switches

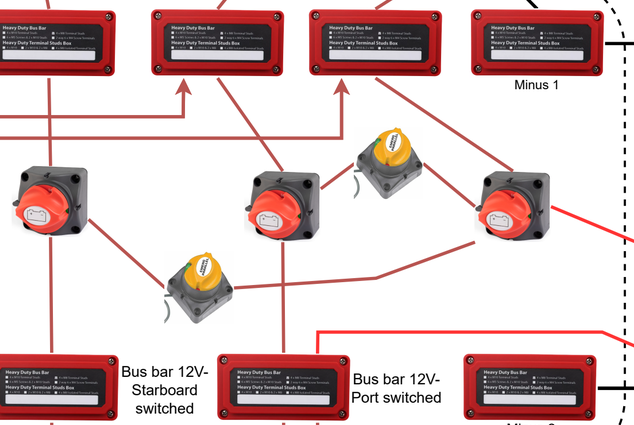
Fuse Box



Répartiteur



Répartiteur



Minus 1

Bus bar 12V-Starboard switched

Bus bar 12V-Port switched

Minus 2

Fuse

Fuse

Switch, BusBar and Coupler Box

#22729 - Battery 12V thermique

Alternator

Alternator



Moteur babord/Port



Moteur Tribord/Starboard



Bus bar +&- Direct plus



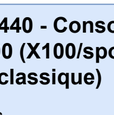
Helm switch or key with remote control switch (exclusive version)

Bus bar +&- Swiched



FB Console

Bus bar +&- Fan



Keybox X100 Therm

Fan cmd

#20488 - Set cooling fan

FB IMS



Engine Starter & Engine water pump & Exhaust water pump

Relay Power ON

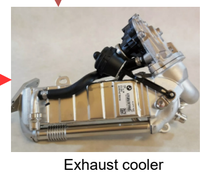
Relay Fan Exhaust

#28399 - KeyBox Engine Version EB2

- CAN
- Supply 12V
- **Cooling Fan Cmd**
- **Fan exhaust Cmd**
- Engine
- **Starter Inboard Cmd**
- AUX 1
- AUX 2

Version Kubota: #31697





Exhaust cooler

Alternateur

Moteur Inboard

#33371 - Engine EB2

- port tension sensor
- port tension starter inboard
- Starter supply (=Power ON)
- Engine cmd (Starter, engine water pump & exhaust water pump)
- J1939
- Tacho output (pas utilisé)

Version Kubota: #39209

#26848 - FuseBox IMS Thermique X100

- Supply 12V
- Supply MD4
- **RTC**
- Direct battery
- Supply MC42
- Fan exhaust (relay)
- **Supply 12V-2**
- Cmd fan exhaust
- Cmd starter inboard
- **Port tension starter inboard**
- Starter supply (relay)

Faisceau option:

#22368 - Guindeau Ancre X100

#36090 - Faisceau Toilette x100

#37489 - Fusion BB100

#22437 - Faisceau table

#34798 - Fusebox RGB X100

GND

12V

230V AC

Lien avec un plan faisceau (RapidHarness)

Thermique

12V (hors Mercury)
Type de bateau: X100 **Thermique**
Author: Arthur Echivard
Date: 19/11/2025
indice: ind A

#33371 - Engine EB2

- port tension sensor
- port tension starter inboard
- Starter supply (=Power ON)
- Engine cmd (Starter, engine water pump & exhaust water pump)
- J1939
- Tacho output (pas utilisé)

Version Kubota: #39209

#26848 - FuseBox IMS Thermique X100 *(FB IMS)*

- Supply 12V
- Supply MD4
- **RTC**
- Direct battery
- Supply MC42
- Fan exhaust (relay)
- **Supply 12V-2**
- Cmd fan exhaust
- Cmd starter inboard
- **Port tension starter inboard**
- Starter supply (relay)

#28399 - KeyBox Engine Version EB2 *(Keybox)*

- CAN
- Supply 12V
- **Cooling Fan Cmd**
- **Fan exhaust Cmd**
- Engine
- **Starter Inboard Cmd**
- AUX 1
- AUX 2

Version Kubota: #31697

#33530 - MD4 - X100

- Buzzer
- **ID TAG 1**
- Joystick_Left
- Joystick_Right
- Supply MD4 12V
- **RTC**
- CAN Main
- G12
- Wifi
- CAN Garmin **(commuter)**

IMS

FB IMS

Power ON relay ("START" *on FB IMS drawing*)

Fan exhaust relay

Fan exhaust (FAN_Exhaust_Cmd K3)

Keybox X100 Therm

Power ON ("Starter_inboard_cmd" on drawing/ K4 *IQAN* & **FH5** *in keybox*)

Starter ("Engine_Cmd"/ K5 & **FH4**)

Engine water pump (Engine Cmd K6)

Exhaust water pump (Engine Cmd K7)

Fan rear & Fan super (FAN_rear_Cmd K1&K2)

#20488 - Set cooling fan

Exhaust cooler

Sensor: T Exhaust coolant temp (Exhaust temp)

Sensor: RPM, T_huile & T_water (J1939)

Inboard Engine

12V-X100 Therm *(lien sur drawio)*

Commande IMS

MD4

BUS CAN Main

ID Tag 0

G12

#22679 - MC42_X100 *(MC42)*

BUS CAN Drive (keybox)

MC42

Tension Port

Battery port

J1939

Temp Oil

Pump state

RP1&RP2
LP1&LP2

Right pump

Left pump

Hydro pump

- EV By pass MC41
- EV Down MC42
- EV unlock MC43
- EV lock MC59
- EV tension MC58
- EV up MC57
- Brake MC44

Locking pressure

Track pressure

Oil level

Manifold block

Console

Garmin

Fusebox Console

Fusible 3A

Alim NMEA2000

NMEA 2000

VHF

Transpondeur AIS800

Autres accessoires Garmin

Digital

Analogique

CAN/NMEA2000

12V

Link with a harness drawing (RapidHarness)

CAN/Digital/Analogique

Type de bateau: X100 **Thermique**
Author: Arthur Echivard
Date: 19/11/2025
indice: ind A

Mercury

Console



NMEA 2000



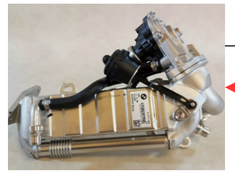
Gateway Mercury <=> NMEA2000



Pompe de direction

Humphree

Bluenav



Exhaust cooler

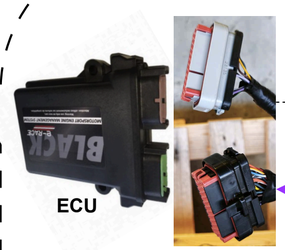
T Exhaust coolant temp mc52 ("exhaust temp MC21" *sur plan MC42*)



Engine inboard EB2

J1939 (CanBus_MC42)

Power ON (violet)

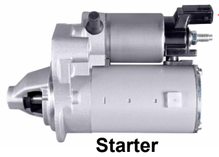


ECU

FuseBox inboard



Starter



Starter



Fuel pump



Alternator



Minus 1

IMS

FB IMS



Power ON relay ("START" *on FB IMS drawing*)



Fan exhaust relay

"T K3 **Fan exhaust**" *sur IQAN*
"Fan Exhaust cmd" *sur plan Keybox*

"T K4 engine **power ON**" *sur IQAN*
("Starter_inboard_cmd" *sur plan Keybox*)

Power On
("starter_supply" *sur plan FB IMS*)



Keybox X100 Therm

"T K1 **Fan Rear**" *on IQAN*
"Fan_Rear_Cmd_cmd K01" *sur plan Keybox*

"T K2 **Fan Super**" *sur IQAN*
"Fan_Rear_Cmd_cmd K02" *sur plan Keybox*

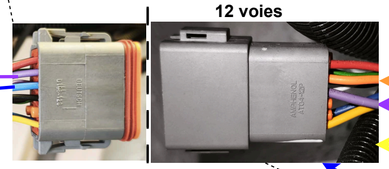
#20488 - Set cooling fan

CAN Keybox



MC42

Connecteur Deutsch 12 voies



Cavity 2
Cavity 3
Cavity 6
Cavity 7
Cavity 9 high &10 low

"T K5 engine **Starter**" *sur IQAN*
"Engine_cmd K05" *sur plan Keybox*

"T K6 engine **water pump**" *sur IQAN*
"Engine_cmd K06" *sur plan Keybox*

"T K7 exhaust **water pump**" *sur IQAN*
"Engine_cmd K07" *sur plan Keybox*

J1939 ("CanBus_MC42" *sur plan*)

port tension sensor

Supply inboard 35mm²



Bus bar 12V- Port switched





Bus bar 12V- service direct



Bus bar +&- Fan

#28399 - KeyBox Engine Version EB2 *(Keybox)*

- CAN
- Supply 12V
- **Cooling Fan Cmd**
- **Fan exhaust Cmd**
- Engine
- **Starter Inboard Cmd**
- AUX 1
- AUX 2

Version Kubota: #31697

#26848 - FuseBox IMS Thermique X100 *(FB IMS)*

- Supply 12V
- Supply MD4
- **RTC**
- Direct battery
- Supply MC42
- Fan exhaust (relay)
- **Supply 12V-2**
- Cmd fan exhaust
- Cmd starter inboard
- **Port tension starter inboard**
- Starter supply (relay)

#33371 - Engine EB2

- port tension sensor
- port tension starter inboard
- Starter supply (=Power ON)
- Engine cmd (Starter, engine water pump & exhaust water pump)
- J1939
- Tacho output (pas utilisé)

Version Kubota: #39209

#20488 - Set cooling fan

#22679 - MC42_X100 *(MC42)*

GND

12V

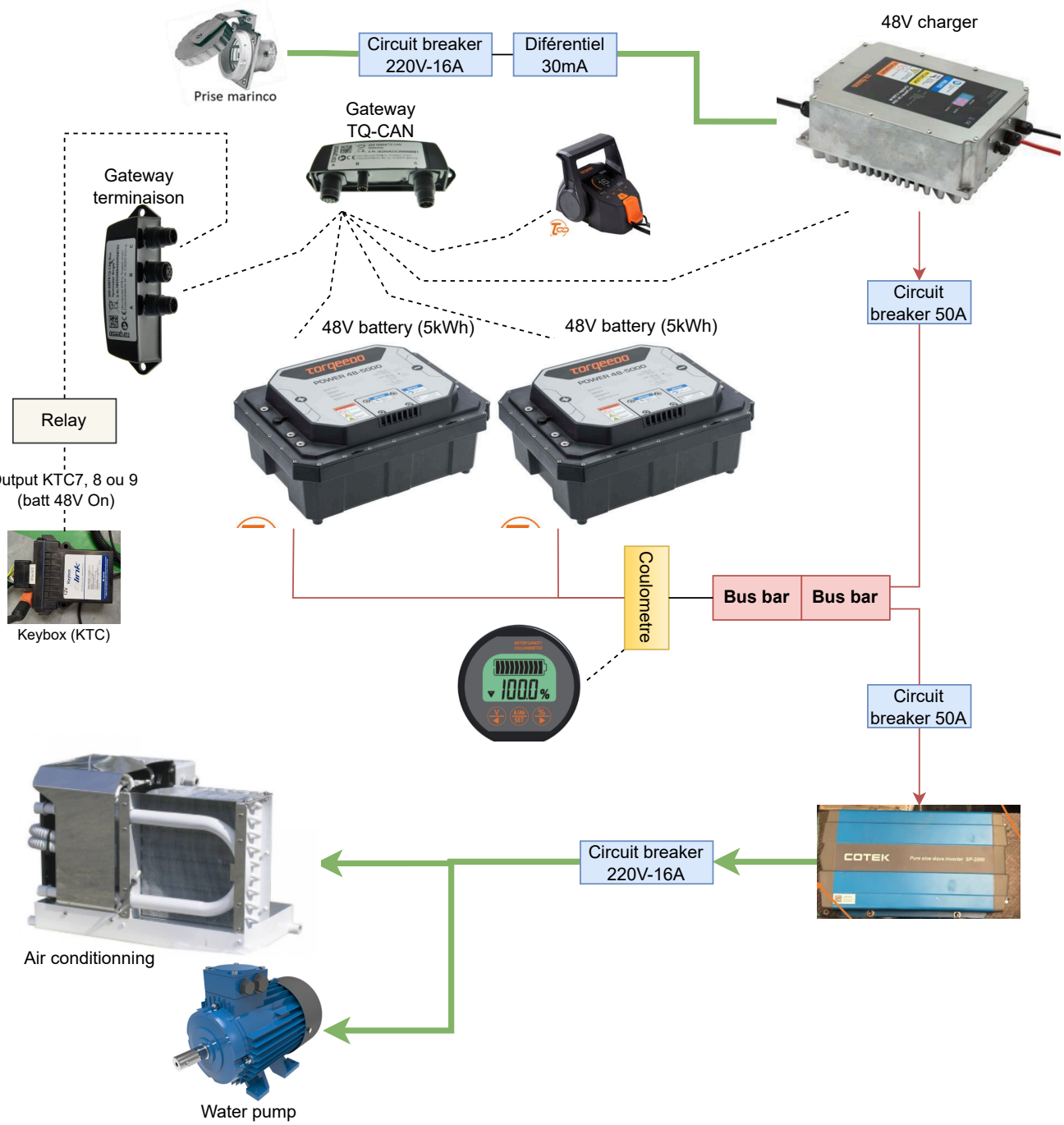
230V AC

Paramètres pour tester les fonctionnalités de l'EB2 (dans "Master Features" & "Master Thermic Cooling"):

- **ECU Engine on**: Active le power ON de l'ECU (K4 sur IQAN et FH5 dans la keybox)
- **ECU Engine priming:** Active et désactive le power ON de l'ECU toutes les 6s
- **Manual Start Engine:** Active le starter (K5 sur IQAN et FH4 dans la keybox)
- **Exhaust Fan Test:** Active K3
- **Aft Fan Low Test:** Active K1
- **Aft Fan Max Test:** Active K2
- **Force Engine Pump:** Active K6
- **Force Exhaust pump:** Active K7

Pour plus d'information **voir "plan faisceau EB2"**, lien: *Dropbox/design/81 BE/04-Equipe BE/ Arthur ECHIVARD/2- Datasheet/Thermique/EB2*

12V EB2
Type de bateau: X100 **Thermique**
Author: Arthur Echivard
Date: 21/11/2025
indice: ind A



Data/CAN

400V

12/24/48V

Water

230V AC

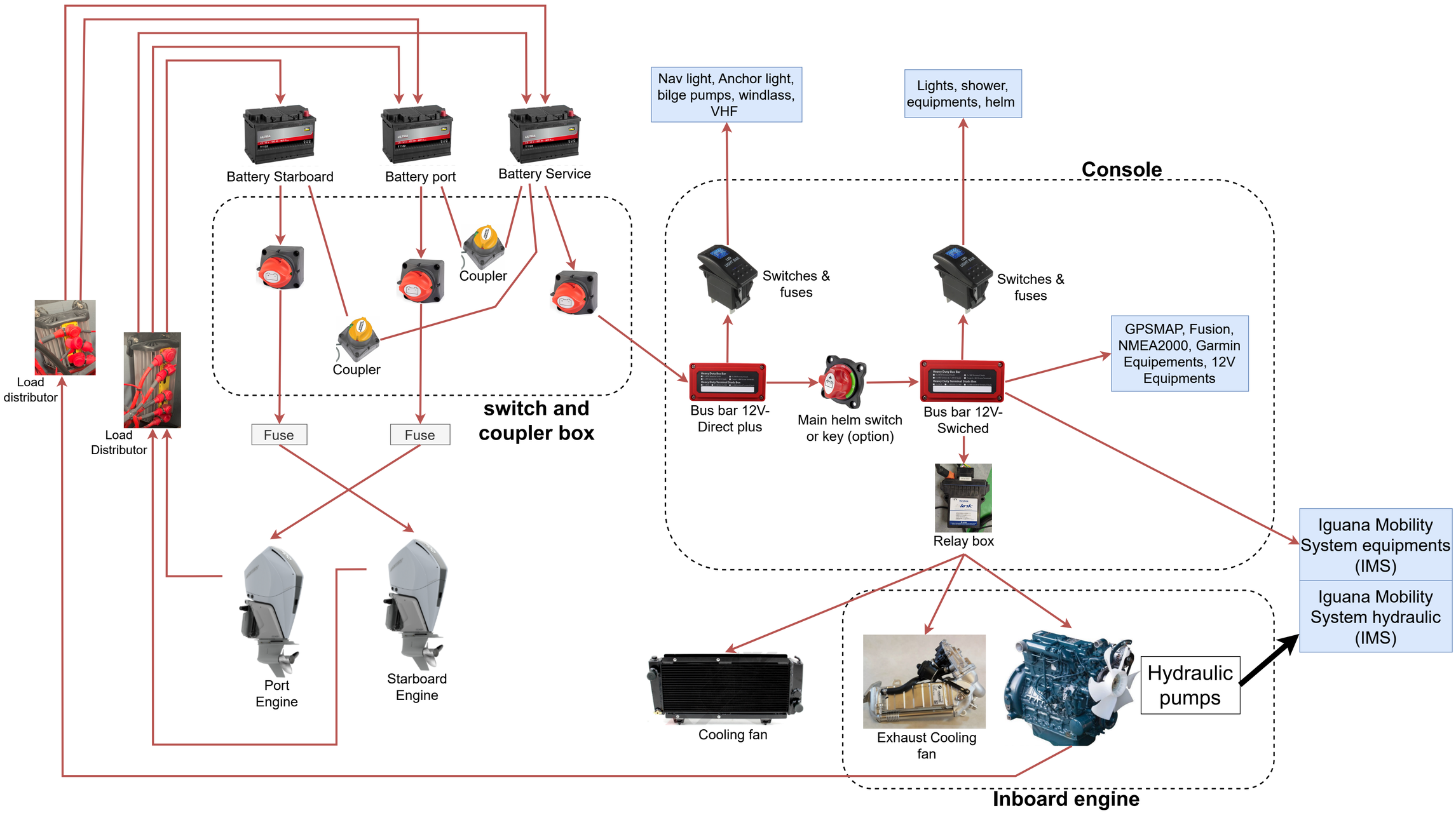
Climatisation V6

Type de bateau: ALL

Author: Arthur Echivard

Date: 20/02/2025

indice: ind A



12V
Type de bateau: X100 **Thermique**
Author: Arthur Echivard
Date: 10/09/2025
indice: ind A

Battery Starboard

Battery port

Battery Service

Coupler

Coupler

switch and coupler box

Load Distributor

Fuse

Fuse

Port Engine

Starboard Engine

Nav light, Anchor light, bilge pumps, windlass, VHF

Lights, shower, equipments, helm

Console

GPSMAP, Fusion, NMEA2000, Garmin Equipements, 12V Equipments

Switch IMS

Switches & fuses

Switches & fuses

Bus bar 12V-Direct plus

Main helm switch or key (option)

Bus bar 12V-Swiched

IMS latching relay

Bus bar 12V-IMS

Iguana Mobility System equipments (IMS)

12V

400V

Battery 400V

DC-DC 400-12V

AC charger

Breaker

Prise marinco

Inverter

Pump inverter

Inverter

Port track **motor**

Hydraulic **pump**

Stbd track **motor**

Iguana Mobility System hydraulic (IMS)

400V system

400V DC

230V AC

12V DC

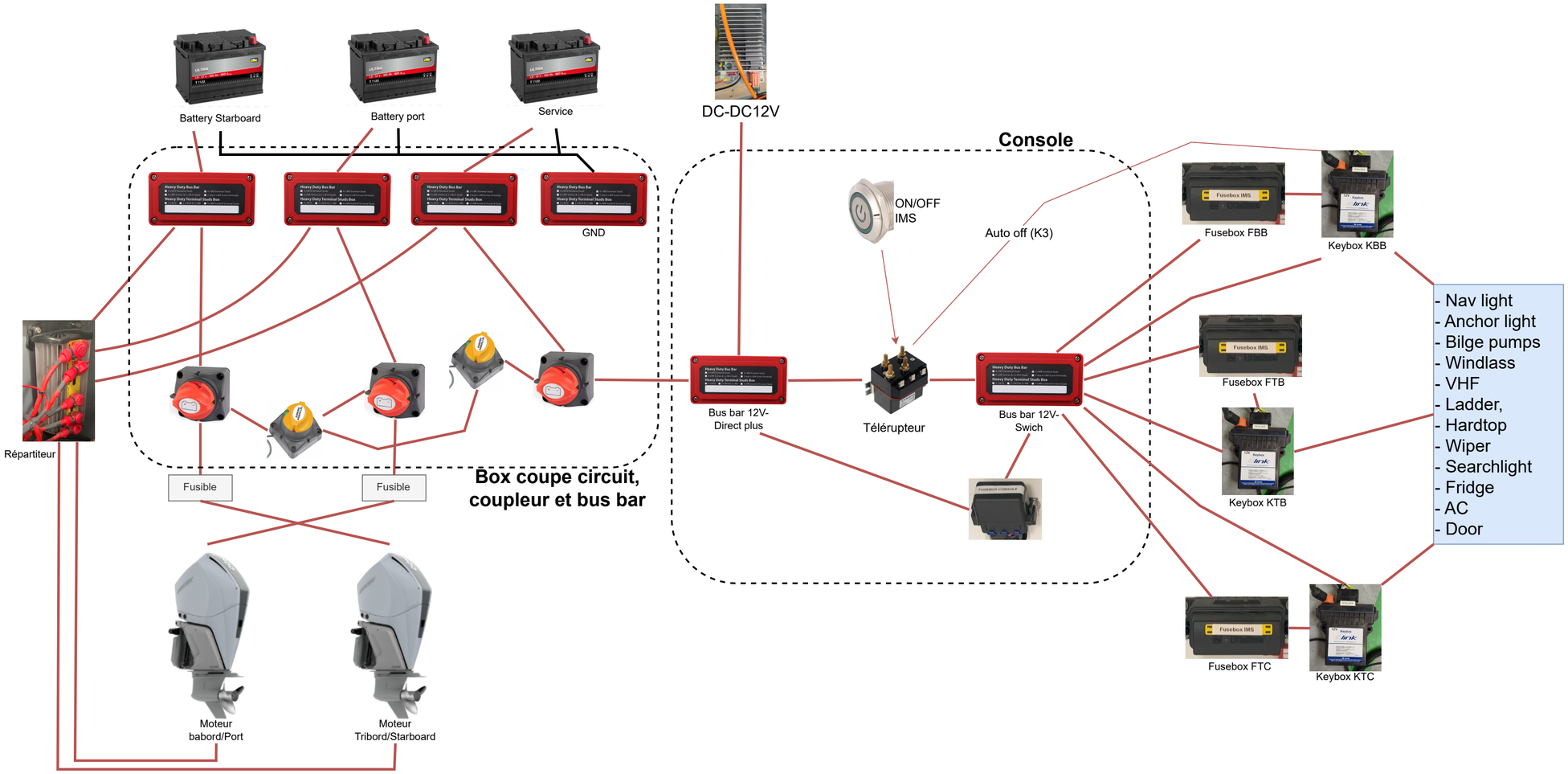
12V & 400V

Type de bateau: X100 **Electric**

Author: Arthur Echivard

Date: 10/09/2025

indice: ind A



12V & 24V (hors Mercury)_pas fini
Type de bateau: X100 **Electrique**
Author: Arthur Echivard
Date: 18/09/2025
indice: ind B

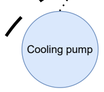
Commuter

Équipements

- Ladder
- Hardtop
- GPSMAP
- Audio
- VHF
- lights
- fridge
- anchor
- bilge
- shower


KTB


KTC


Cooling pump


FAN mk E


DC-DC12V


KBB


Télérupteur IMS


230V AC charger

Battery 400V


Sevcon Gen4 inverter


Sevcon Gen4 inverter


Sevcon HVLP inverter

BUS CAN Drive

IMS

Commande IMS


MD4





BUS CAN Main

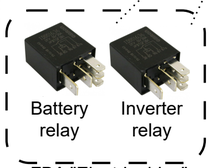

G12

BUS CAN Drive


MC42

IMS ON

IMS ON


Battery relay

Inverter relay

FBA "Electric drive"

Distributeur hydraulique
- EV By pass MC41
- EV Down MC42
- EV unlock MC43
- EV lock MC59
- EV tension MC58
- EV up MC57


Oil level


Locking pressure


Track pressure


T hydraulic pump statue

T oil Temp

Console

Garmin






Fusebox Console

Alim NMEA2000

NMEA 2000


VHF

GPS





Data

400V

12V

Water

230V AC

CAN & Commande12V_Pas fini	
Type de bateau:	Commuter
Author:	Arthur Echivard
Date:	20/02/2025
indice:	ind A

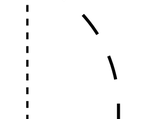
Mercury

Console



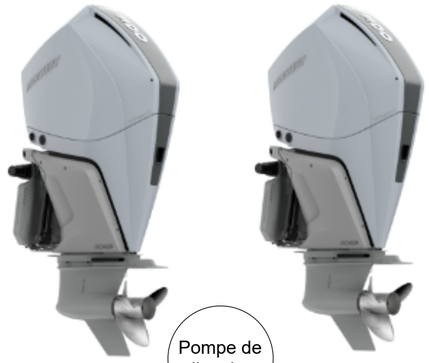






NMEA 2000





Pompe de direction

Humphree

Bluenav